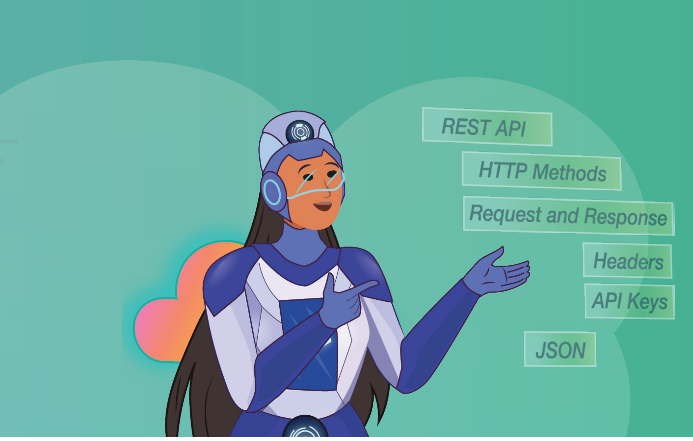


Weather API

03

Hello Techna! I am creating a website and want it in multiple languages. However, I am unable to implement the language translation feature to it.

Siya, you can use APIs that provide such language translation service. APIs provide seamless functionalities of translation. Your audience can view the website content in their preferred language.

Imagine you are travelling to a foreign country, and you do not know the native language. So, to interact with the locals, you will use a guidebook or a translation. Relating to this example, let us understand how computers interact with different software applications.

Computer software need to communicate with other software and web applications for sharing and accessing resources and services. It is similar to how humans communicate with each other. Software or Web applications communicate through APIs.

3. Weather API

What is an API?

Application Programming Interface (API) can be defined as the channel or medium for communication between software or web applications. It provides a secure and standardized way for applications to work with each other and deliver information or services without any user intervention.

One of the main advantages of working with APIs is that they perform heavy operations in the background. The user only needs to request the API for the service and utilize it.

For example, if you are building an application and want to integrate 'biometrics' for authentication, you may simply connect with an API that provides biometric authentication services rather than creating one on your own.

Understanding APIs

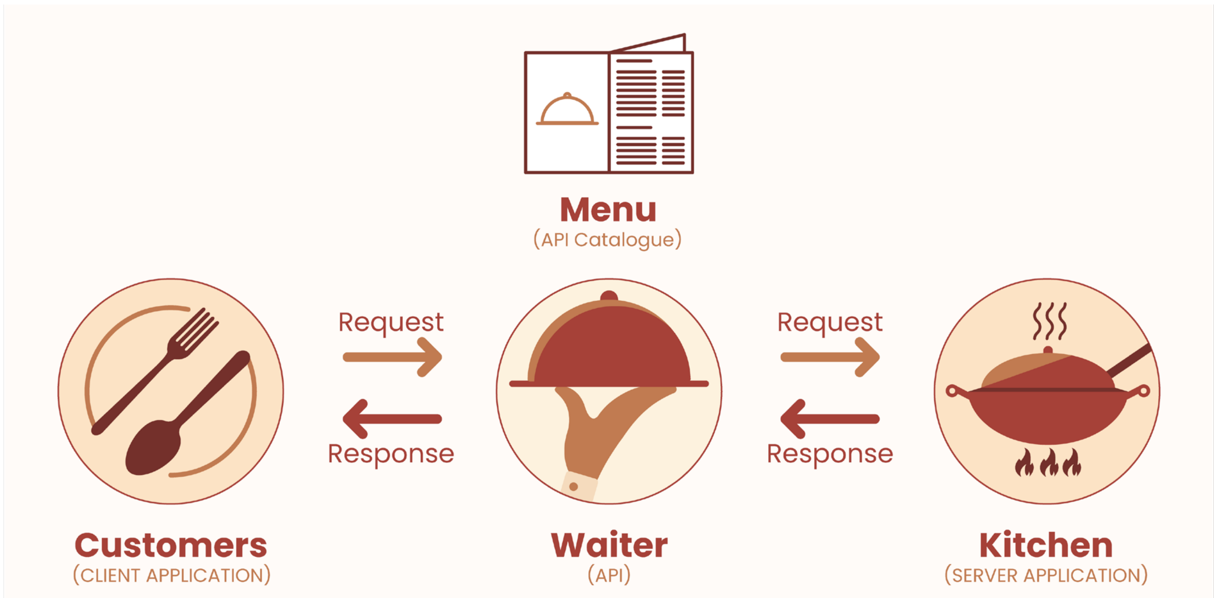
Suppose we want to build an application that includes a map of a business location or displays a list of the latest tweets tweeted by the organization. We cannot directly access the servers of Google or Twitter and fetch the data, but we may use 'Google Maps API' and 'Twitter API'. These APIs allow users to get those data (maps and tweets) and integrate them into our application.

A real-life example of API utilization is 'Uber'. It uses the Google Maps API to implement maps on its app, rather than building a location mapping platform.

There are various software and web applications available on the internet that provide a variety of services, such as processing online payments, displaying maps, embedding YouTube videos, and so on. So, if a user has any such requirements, instead of developing those applications from scratch, the user can simply connect to their previously created APIs and use their services instead. In simple words, an **API** is something that lets a user connect to a remote server and utilizes their services.

How do APIs Work?

Before we understand how APIs work, let us consider a simple scenario.



You are hungry and decide to visit a restaurant. You go through the menu and ask the waiter for a burger (let's say). The waiter takes your order to the chef/cook, who then prepares your burger and serves it to you.

In this comparison, the customer is the application that requests a service, the waiter is the API that will handle the request, and the chef is the server application that will process your request and deliver the result to you. Just like a restaurant has a distinct menu, an API also offers a certain range of services. A user request must be limited to those services.

When we want to use an API, we need to know how to ask for things in a way that the API can understand. It is like ordering your food in a language that the waiter speaks. The language that the API understands is called the 'API protocol', which is a set of rules for how to ask for things and how to receive them.

The following points provide a high-level overview of how APIs work -

- **Sending Request** - A software or a web application first sends a 'request' to the API server, asking for a particular service or resource. A 'request' is simply a message from a user(client) to a server that contains information about what the web server requires.
- **Processing** - Based on the request received, the API will 'process' the request, following the rules already defined in the API.
- **Sending Response** - Once the request is processed, the API sends a 'response' (containing the requested data) back to the application. The API also handles the task of converting the data into a format, that is understandable by the requesting application.
- **Handling Response** - The application can now use the response (which contains the requested data) in accordance with its specifications.

Advantages of using API

There are several advantages of using an API, as mentioned below:

a. APIs allow different types of applications to communicate and share resources or data with one another.

b. With APIs, developers are not required to code the functionality from scratch because it is already coded in the API. This leads to faster and more efficient application development.

c. APIs make it simple to scale applications to handle increased traffic.

d. APIs transfer data in a secured format, preventing sensitive information from being exposed or tampered with.

Real World Examples of API

Following are a few APIs that provide various helpful services -

- **Google Maps API** - This API provides access to Google's mapping and location data. Developers can use the API to add interactive maps and location-based features to their applications.
- **Twitter API** - Using this API, we can access and retrieve tweets, monitor hashtags, post tweets, etc. in real-time.
- **NASA API** - This API is provided by 'NASA' and can be used to retrieve data on Asteroids, Earth Imagery, Mars Rover Photos, and many more.
- **PayPal API** - This API enables developers to integrate payment processing service into their applications.
- **Amazon S3 API** - Using this API, we can store and retrieve data stored on Amazon's cloud storage service, 'Amazon S3'.
- **Weather API** - This API provides weather reports for any country/city in the world.

API Key Components

Before getting started with APIs, it is better to get familiar with some of the concepts/definitions associated with APIs.

- **API Call** - An 'API call' refers to a request sent by a client computer for some service or data. (Analogy – Ordering a dish you want to have). This request specifies which data to retrieve or which operation to perform.

3. Weather API

- **API Endpoints** - 'API Endpoints' refer to specific URLs that can fetch a particular service or data from the API. When a client application requests API, it specifies the endpoint to which it wishes to connect, as well as any parameters or data that the request requires.

Example - If an API provides weather details, it might have endpoints such as:

- **/weather/forecast** - for accessing weather details.
- **/weather/tomorrow** - for accessing weather details of the next day.
- **/weather/current** - for accessing the present weather details.

Request Methods - While making an API call, the client must specify a request method that defines what action the client wants to perform. It could be any one of the following –

- **GET:** To retrieve/fetch a service or data from the API server. It is one of the most used requests.
- **POST:** To send data or information to the API server. It is frequently used to submit HTML Form data.
- **PUT:** To edit or update existing data or information on the API server.
- **DELETE:** To remove the existing data from the API server.

API Key - An 'API Key' is a unique identifier that is used to authenticate API calls. Every user who interacts with an API is assigned a unique API key.

- **API Documentation** - Each API has its own documentation that offers advice on how to effectively use and integrate the API properly. It is important to read any API documentation because each API has a unique set of instructions for how to utilize it.

Exercise

Fill in the blanks.

1. 'API' stands for ____________________.

2. To fetch a service or data from an API, the ____________________ request method is used.

Tick mark ☑ the correct answer

3. The most widely used API for providing location and mapping data is

a. Google Maps API ☐ **b.** Twitter Maps API ☐
c. Stripe API ☐ **d.** Paypal API ☐

4. Which of the following is the correct definition of an API Endpoint?

a. Unique URLs that fetch a particular service or retrieve data from an API Server. ☐
b. Unique Identifier that authenticates requests made to an API. ☐
c. Refers to the request sent by a client to an API. ☐
d. Contains instructions on how to use an API ☐

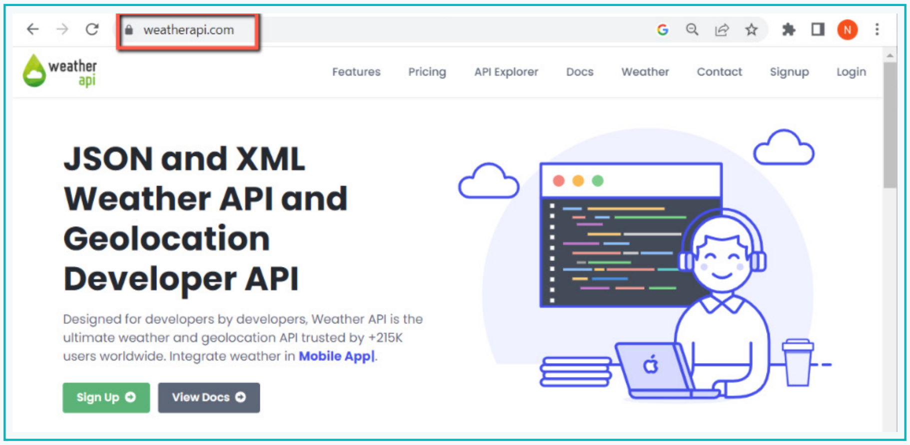
5. Which API provides payment processing services?

a. Amazon API ☐ **b.** Google Maps API ☐
c. Paypal API ☐ **d.** NASA API ☐

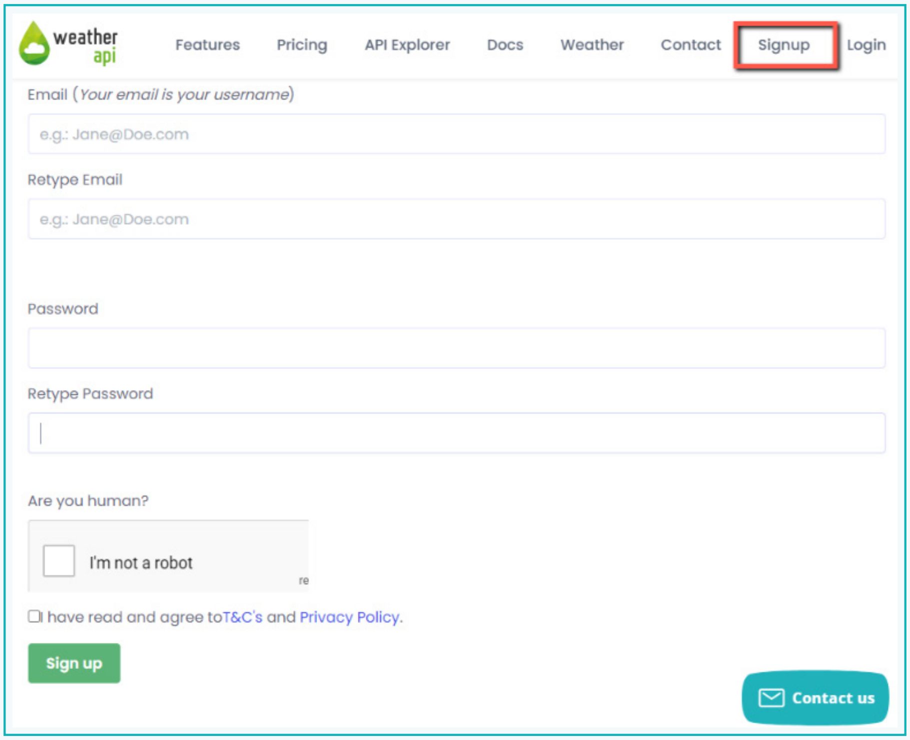
Activity

Step 1: Use the Weather API

1. To access the API, we will visit the website https://www.weatherapi.com/

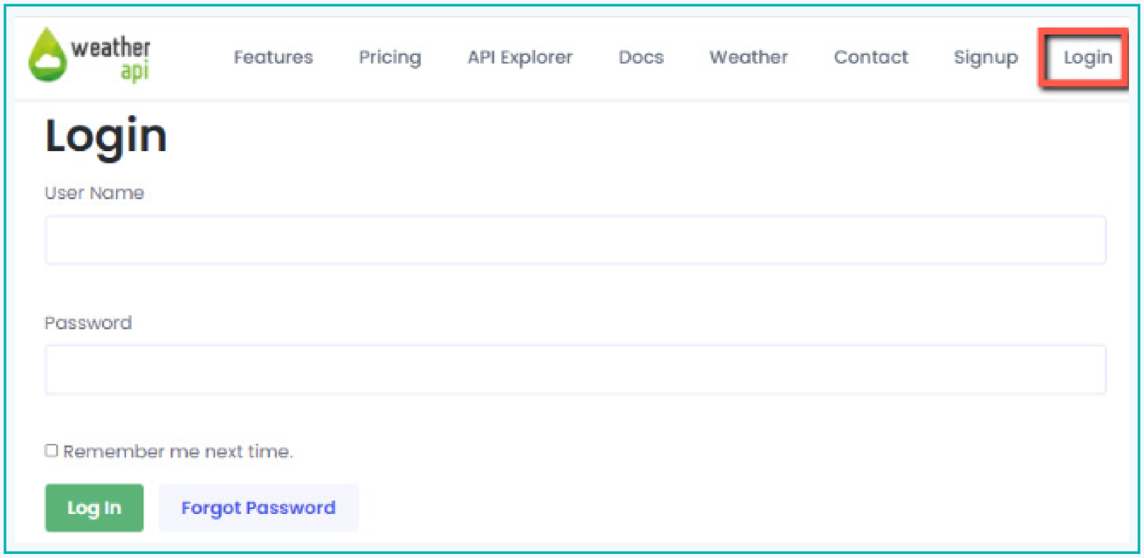


2. Once on their website, click on the ‘Sign Up’ button. Provide the necessary details and submit the form.

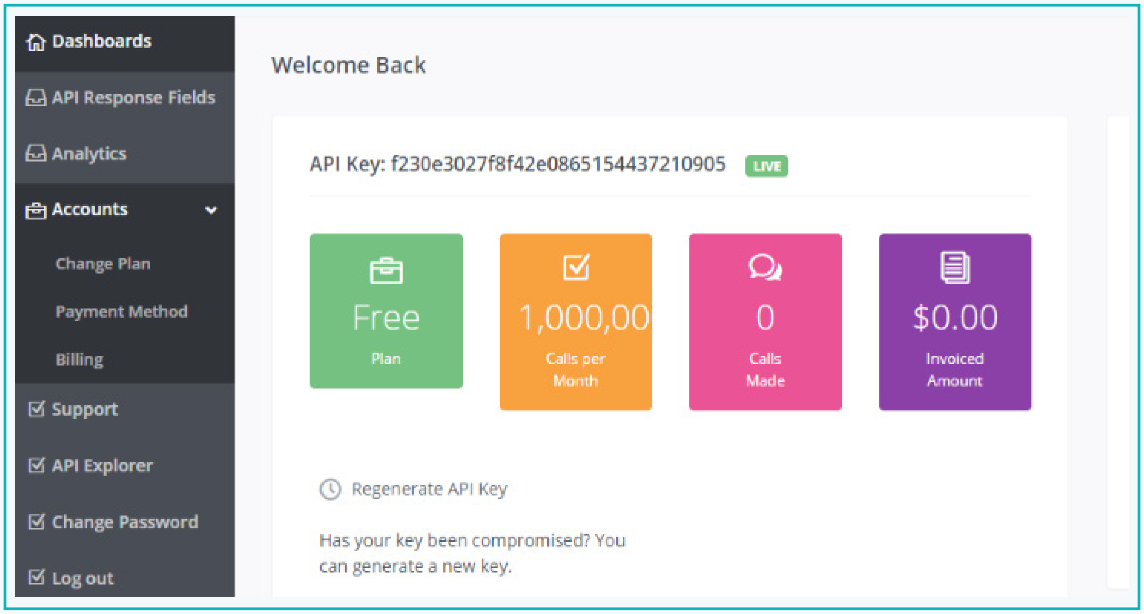


Activity

3. Once you are done with the registration, click the 'Login' button and enter your email and password to log in.



4. Once you are successfully logged in, you will be directed to your Dashboard. The 'Dashboard' is a webpage, where all the details of the API and other related functionalities are provided.



Activity

You will notice that you have been given an API key. The 'API Key' is a unique identifier and is auto-generated. Every user who registers will be assigned a unique API key. Its purpose is to uniquely identify all users attempting to connect with it.

5 Copy the API code and keep it with you (you can store it in a normal text file or write it down on a piece of paper). Next, click on the 'API Explorer' link available on the Sidebar.

6 In the new window,

a. Enter the API key in the textbox labelled 'Your API Key'.

b. Next, click on the 'Protocol' dropdown box, and select 'HTTPS'. This will encrypt our connection to the API, making it secure.

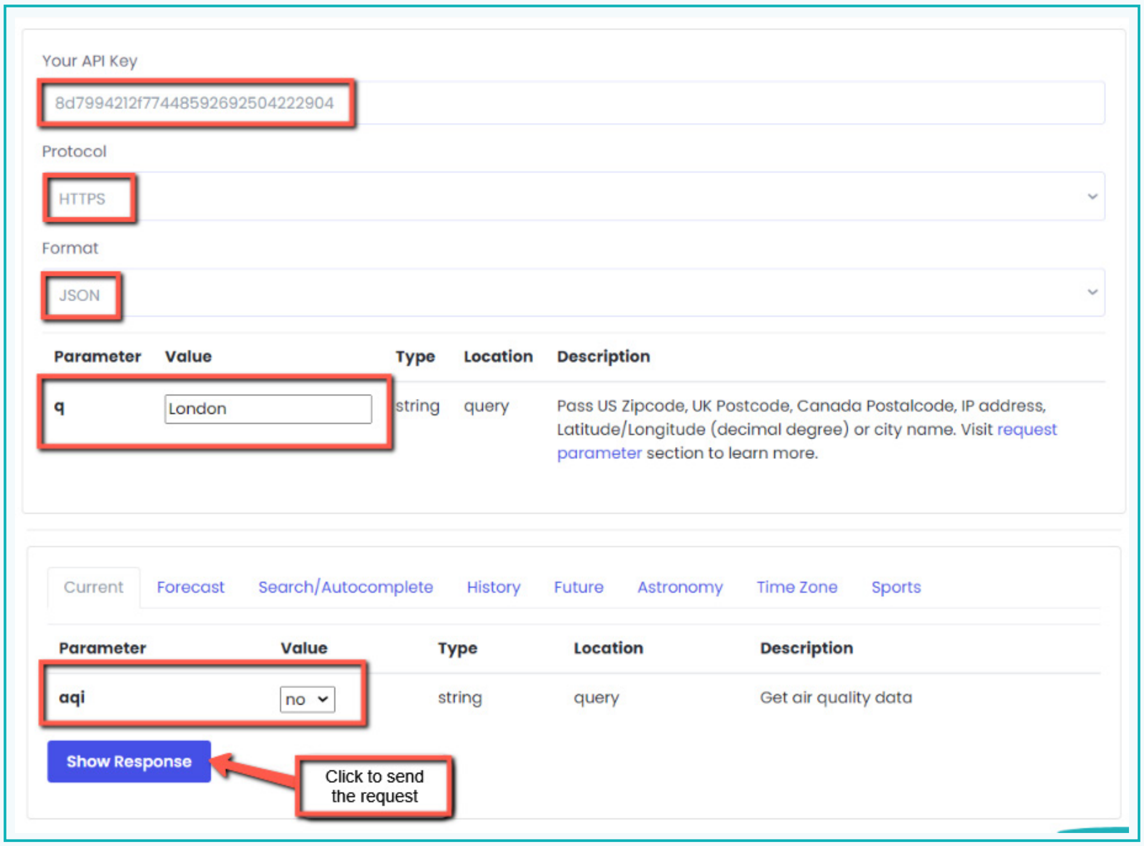
c. For the 'Format' dropdown box, keep it JSON. This indicates that the answer from the API server will be in JSON format.

d. In the textbox containing the value "London", simply enter a state/city name whose weather details you want to check or keep the default value.

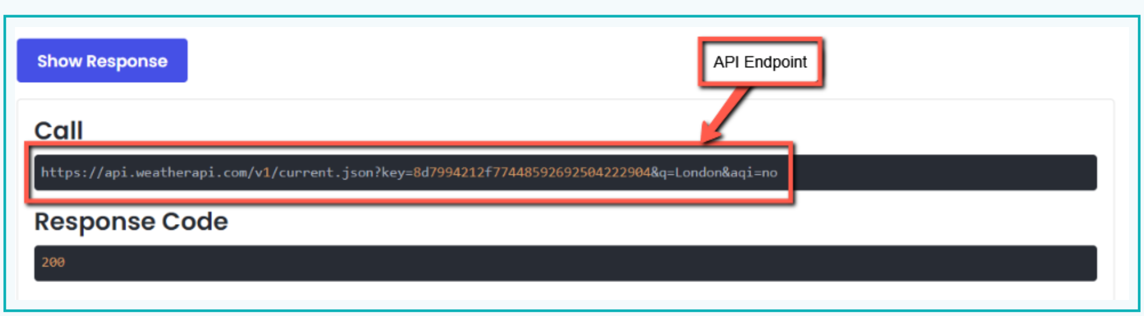
e. For the 'aqi' parameter, keep the value as 'no'. This parameter specifies whether we want to include the air quality information in the response or not.

f. Finally, click on the 'Show Response' button.

Activity



7. By clicking the ‘Show Response’ button, we are requesting the Weather API to return all weather-related data based on the input we provided. The API will return a response with the weather information as well as an ‘API Endpoint’.



Activity

API Endpoint

The URL given below is generated by the API and is referred to as the ‘API Endpoint’ and it contains the ‘base address’ and all the necessary ‘query parameters’ required to generate the output. Let us analyse the URL and understand each parameter’s purpose.

The “https://api.weatherapi.com/v1/current.json” is the base address of the Weather API Service.

The query parameters specified in the endpoint are as follows -

key=8d7994212f77448592692504222904	This parameter specifies the API key.
q=London	This parameter is used to specify the name of the location for which weather information will be provided. Here, it is set to ‘London’.
aqi = no	This parameter specifies whether or not air quality data should be included in the response. In this case, it is set to “no”, which means that air quality data will not be included.

Activity

8. We can see that, after clicking on the ‘Show Response’ button, we got a response from the API Server. The response is divided into 2 categories – ‘Response Header’ and ‘Response Body’.

Response Headers


```
{
  "Transfer-Encoding": "chunked",
  "Connection": "keep-alive",
  "Vary": "Accept-Encoding",
  "CDN-PullZone": "93447",
  "CDN-Uid": "8fa3a04a-75d9-4707-8056-b7b33c8ac7fe",
  "CDN-RequestCountryCode": "GB",
  "CDN-ProxyVer": "1.03",
  "CDN-RequestPullSuccess": "True",
  "CDN-RequestPullCode": "200",
  "CDN-CachedAt": "07/06/2023 05:04:47",
  "CDN-EdgeStorageId": "1072",
  "CDN-Status": "200",
  "CDN-RequestId": "3be39418c67bf1d0066cfd0a08f7b3f1",
  "CDN-Cache": "MISS",
  "Cache-Control": "public, max-age=180",
  "Content-Type": "application/json",
  "Date": "Thu, 06 Jul 2023 05:04:47 GMT",
  "Server": "BunnyCDN-FR1-951"
}
```


Response Body


```
{
    "location": {
        "name": "London",
        "region": "City of London, Greater London",
        "country": "United Kingdom",
        "lat": 51.52,
        "lon": -0.11,
        "tz_id": "Europe/London",
        "localtime_epoch": 1688619887,
        "localtime": "2023-07-06 6:04"
    },
```


The information provided in the ‘Response Headers’ section, is not meant for us. This information is used by the browser to know about the status of the response, connection type and other related data.

Activity

The information that we need to be concerned about is provided in the ‘Response Body’. This part of the response contains all the details related to our API Call. In our case, the Response Body contains various weather-related information such as – weather location, temperature, wind speed, humidity, pressure and so on.

Response Body


```
{
    "location": {
        "name": "London",
        "region": "City of London, Greater London",
        "country": "United Kingdom",
        "lat": 51.52,
        "lon": -0.11,
        "tz_id": "Europe/London",
        "localtime_epoch": 1683531121,
        "localtime": "2023-05-08 8:32"
    },
    "current": {
        "last_updated_epoch": 1683531000,
        "last_updated": "2023-05-08 08:30",
        "temp_c": 12.0,
        "temp_f": 53.6,
        "is_day": 1,
        "condition": {
            "text": "Overcast",
            "icon": "//cdn.weatherapi.com/weather/64x64/day/122.png",
            "code": 1009
        },
        "wind_mph": 6.9,
        "wind_kph": 11.2,
        "wind_degree": 220,
        "wind_dir": "SW",
        "pressure_mb": 1020.0,
        "pressure_in": 30.12,
        "precip_mm": 0.0,
        "precip_in": 0.0,
        "humidity": 94,
        "cloud": 25,
        "feelslike_c": 10.4,
        "feelslike_f": 50.7,
        "vis_km": 10.0,
        "vis_miles": 6.0,
        "uv": 3.0,
        "gust_mph": 11.0,
        "gust_kph": 17.6
    }
}
```

Activity

Response Body

'Response Body' contains all the weather-related details based on our API Call. The entire response is divided into 2 JSON objects – 'location' and 'current'. The 'location' object contains details about the place, whose weather details are provided, and the 'current' object contains weather-specific data.

- The data received is formatted in 'key: value' pairs and is quite simple and easy to understand. For example, the **"name": "London"** key-value pair specifies the location name, the **"lat": 51.52** specifies the latitude, the **"temp_c": 12.0** specifies the current temperature in Centigrade, and so on.

> **Note:**
> We can analyse and filter this information with the help of languages such as JavaScript, and display it on a webpage.

JSON basics

As we have already seen, the response provided by the API Server is in JSON format. Let us first define JSON.

- **JSON,** which stands for **JavaScript Object Notation,** is a file format that is used to store and send data to and from web applications.
- Data is stored in a JSON file in key-value pairs, making it easy to use and understand.
- In JSON, string values are always specified within double quotes(" ") and numeric values do not contain any quotes.
- In JSON, objects are created using curly braces { } and arrays using square brackets [ ].

Here are a few examples of data stored in JSON format.

1. The following is a simple JSON object with two 'key-value' pairs.

```
{
    "name": "Nemo",
    "age": 35
}
```

Activity

2 The following is a JSON object that contains yet another JSON object **(address)** within itself.

```
{
    "name": "Nemo",
    "age": 35,
    "address": {
        "street": "123 Main Street",
        "city": "XYZ Town",
        "state": "State ABC",
        "zip": 12345
    }
}
```

3 The following JSON object contains an array **(hobbies)** for storing multiple related values.

```
{
    "name": "Nemo",
    "age": 35,
    "hobbies": [
        "reading",
        "traveling",
        "hiking"
    ]
}
```

In this manner, JSON allows us to structure different types of data, which can subsequently be exchanged between communicating applications.

Exercise

Fill in the blanks.

6. JSON stands for ______________________.

7. In JSON, curly braces { } are used to create ______________________.

Activity

Working with HTTP Clients for API Testing

When working with APIs, it is important to understand a category of applications referred to as ‘HTTP Clients’.

HTTP Clients

An **HTTP Client** is a basic program that allows us to send HTTP requests to servers and handle the responses. With respect to APIs, ‘HTTP Clients’ can be used for testing it by sending requests to the API Server and analysing the response to verify the functionality of the API. Apart from testing, it is also used for other tasks such as consuming, developing and debugging APIs.

There are several API clients available, such as Postman, Insomnia, cURL, etc. The one that we will be using for this activity is Postman.

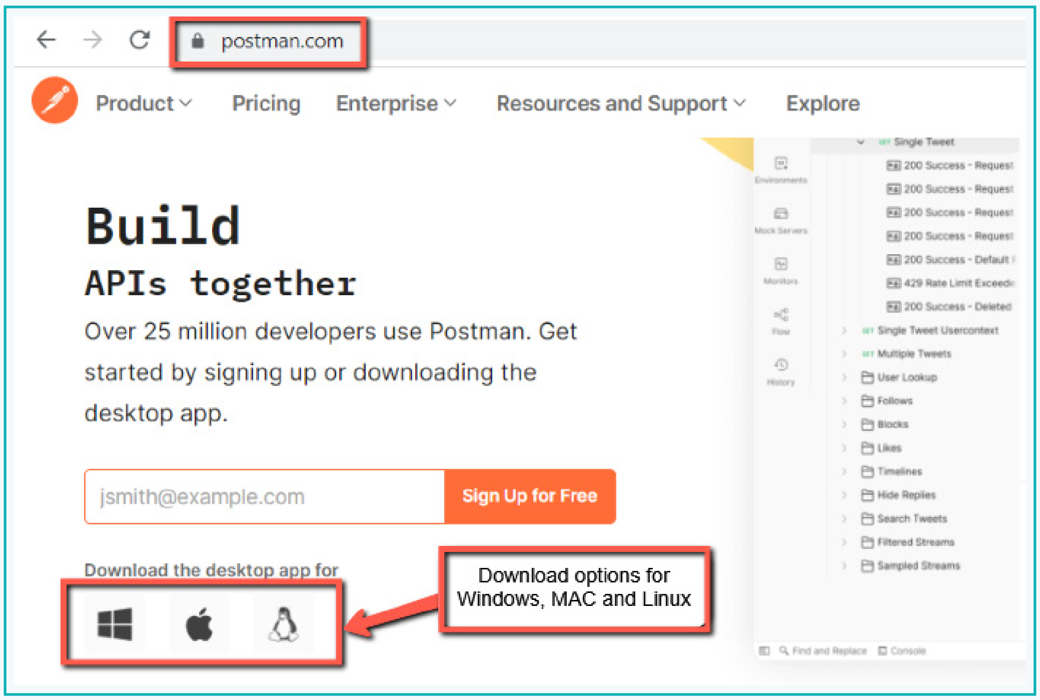
Postman

Postman is a popular API development and testing tool that provides a user-friendly interface for sending HTTP requests, inspecting responses, and testing API endpoints.

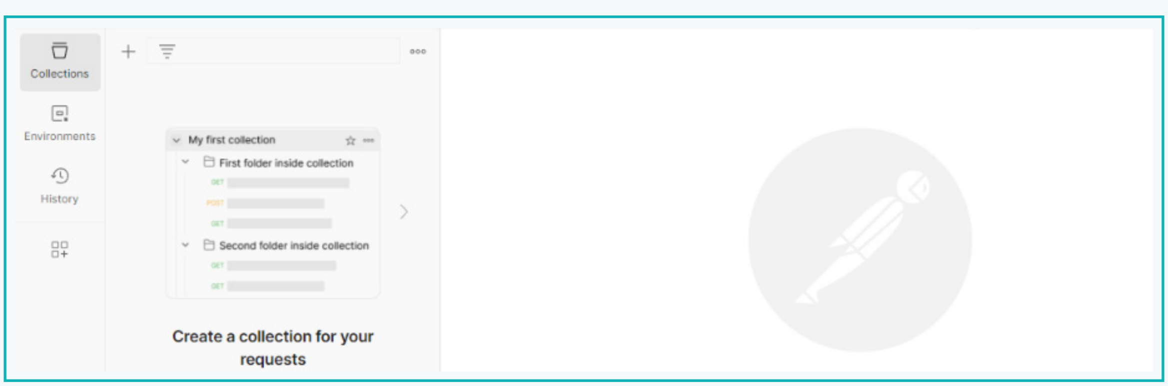
Activity

Step 2: Download the Postman Tool

1. To download Postman, open the site www.postman.com and click on the download icon, available on their homepage. Since we are using Windows, we will click on the first icon.



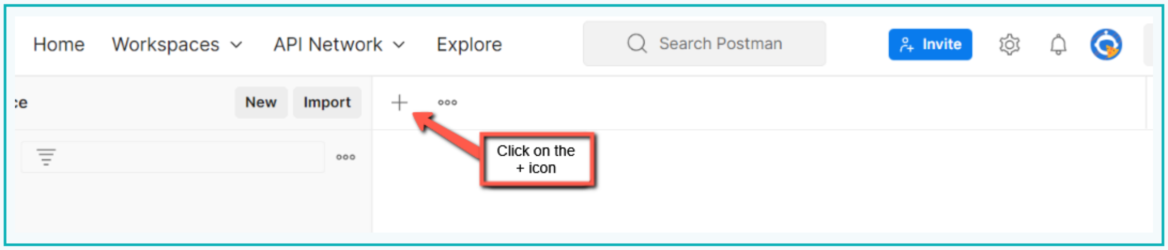
2. Once the file is downloaded, double-click on it to install. Once installed, it will automatically open in your system. The application interface is shown below -



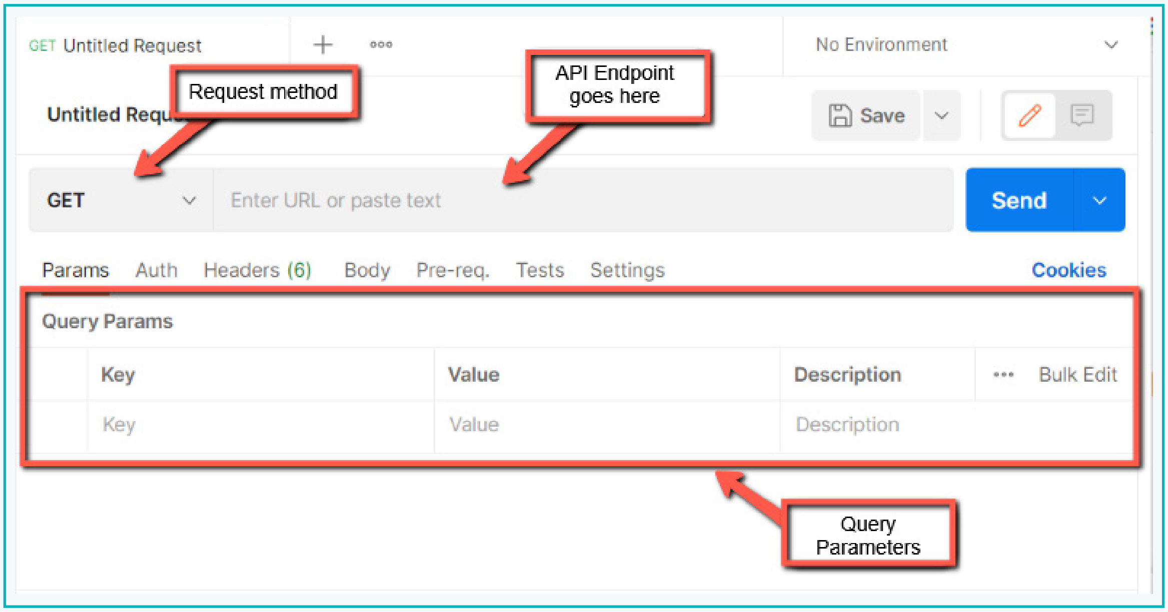
Activity

Step 3: Test the Weather API using Postman

1. To test our API, we need to first open an API Testing window on Postman. This can be done by clicking the '+' icon.

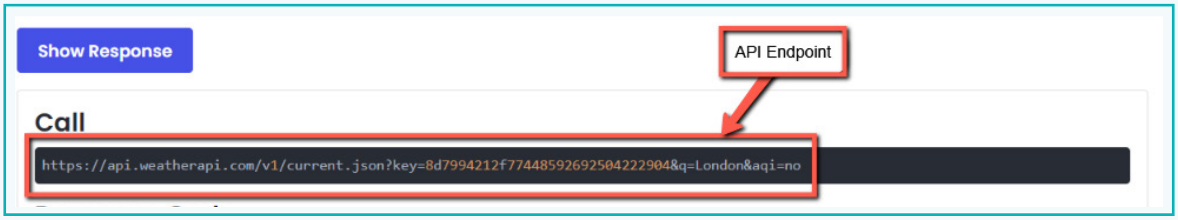


2. This will open a sub window within the application, where we can specify the API Request method, Endpoint, and Query Parameters.

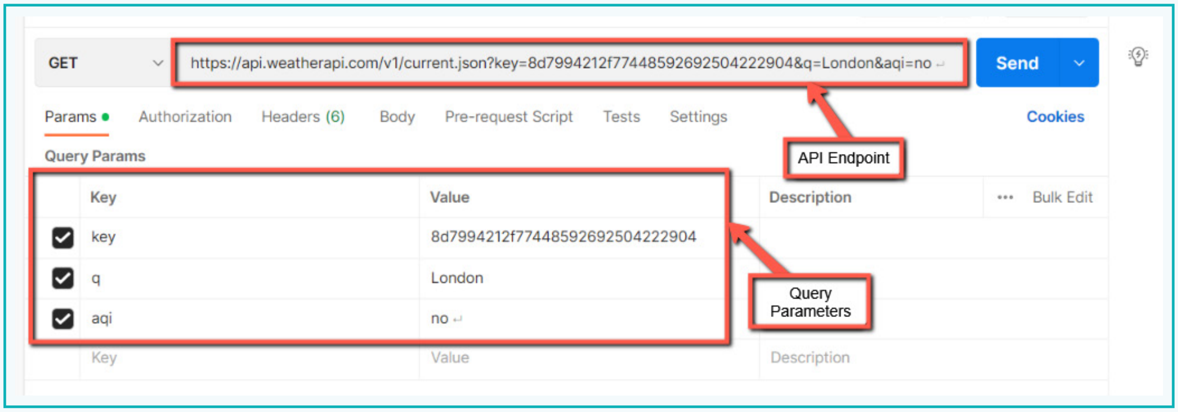


Activity

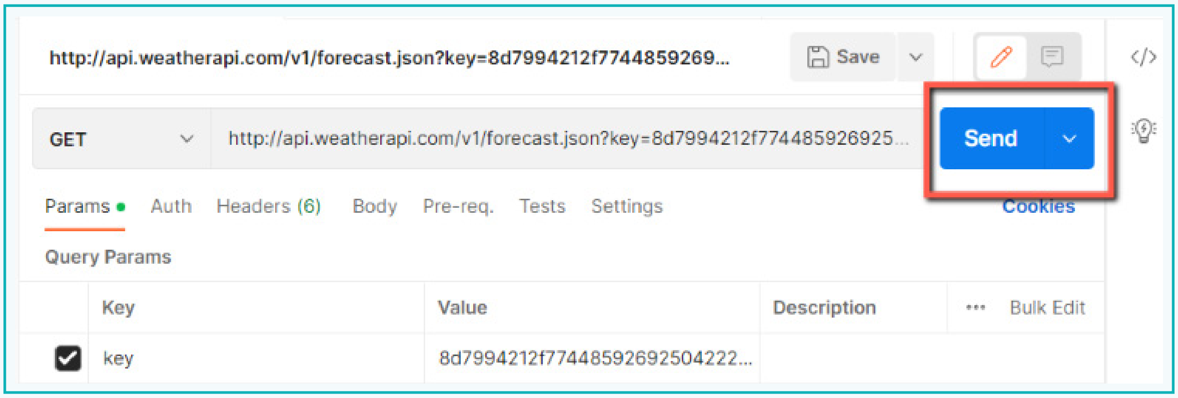
3. To test the Weather API using Postman, we first require the API Endpoint, which is already available to us on the Weather API website.



4. Copy the API Endpoint and paste it into Postman in the space provided. Notice that once the endpoint is provided, Postman can identify the query parameters present in the endpoint and display them in the ‘Query Params’ Section.

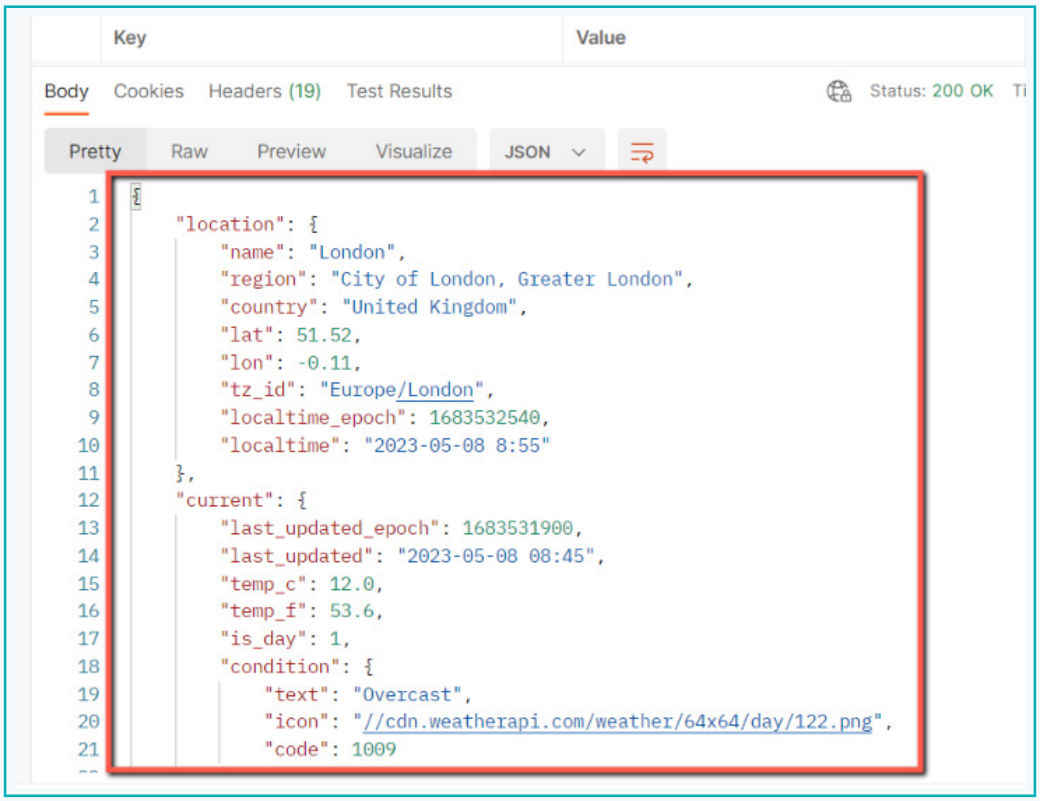


5. Now click on the ‘Send’ button to make Postman send the request to the provided endpoint.



Activity

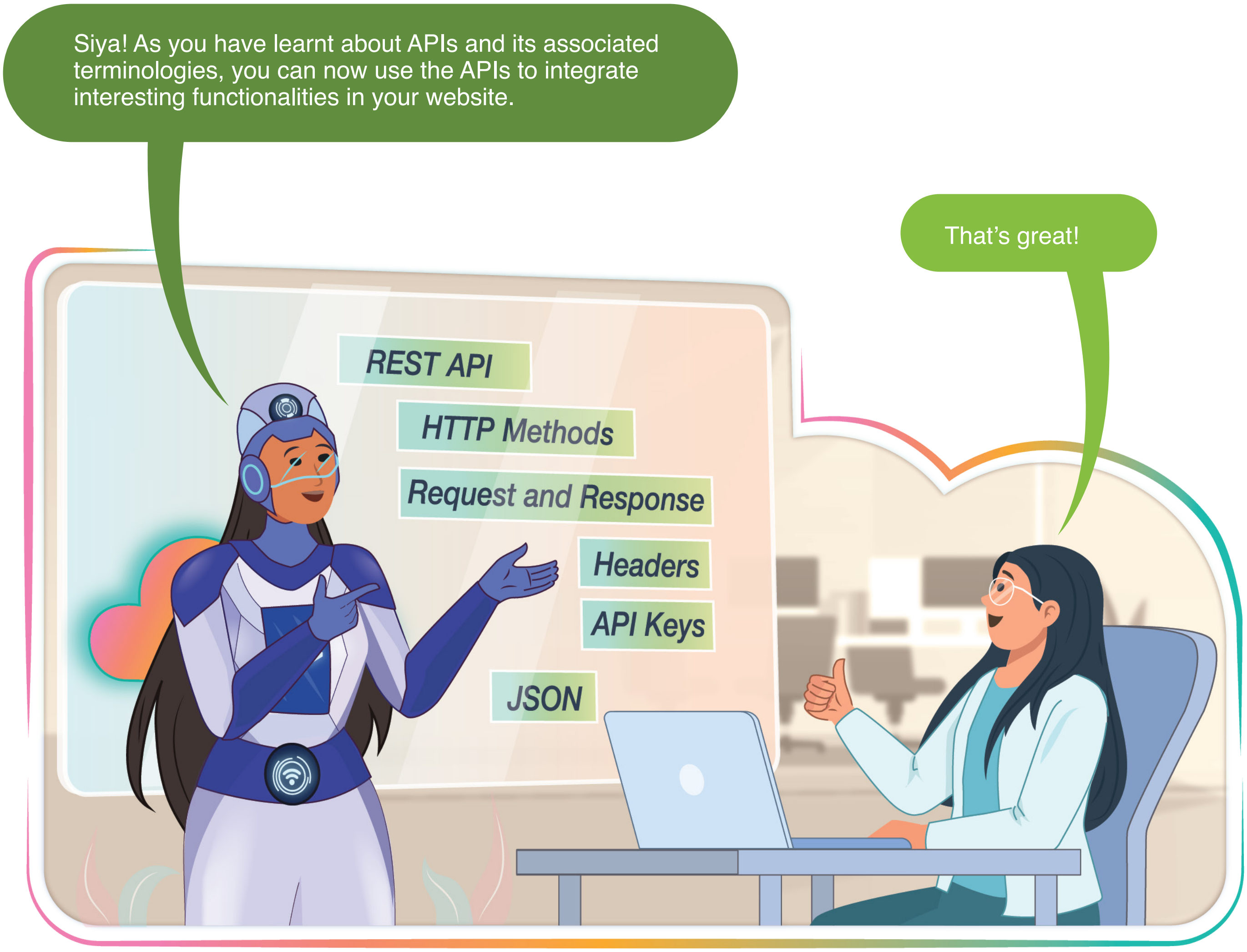
6 Postman will display the response in the same window as soon as the request is sent.



Postman has now provided us with the whole response in a user-friendly format, making it easy to analyse and understand.

So, in this lesson, we got familiar with APIs and how to use it. We also got to know about HTTP Clients such as Postman, that may be used to test APIs.

3. Weather API



Tech Flash

- **API** : It is a software interface that allows interaction between multiple devices for sharing data and/or services. Example – Google Map API, which allows us to embed Google Maps on a webpage.
- **JSON** : A lightweight data-interchange format that is easy for humans to read and while also being simple for machines to parse. Often used for transmitting data between applications.